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- Soal 3 : Segmentasi Pelanggan dan Komposisi Fungsi Harga

Parameter Personalisasi

$N = 22$

$a = 2$

$b = 2$

$k = 5$

$\theta_0 = 70^\circ$

$d = 0.003$

Definisi Himpunan Pelanggan

$A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\} \Rightarrow \{22, 24, 26, 29, 32, 35, 37\}$

(Pelanggan dari media sosial)

$B = \{N+3, N+5, N+7, N+10, N+12, N+18\} \Rightarrow \{25, 27, 29, 32, 34, 40\}$

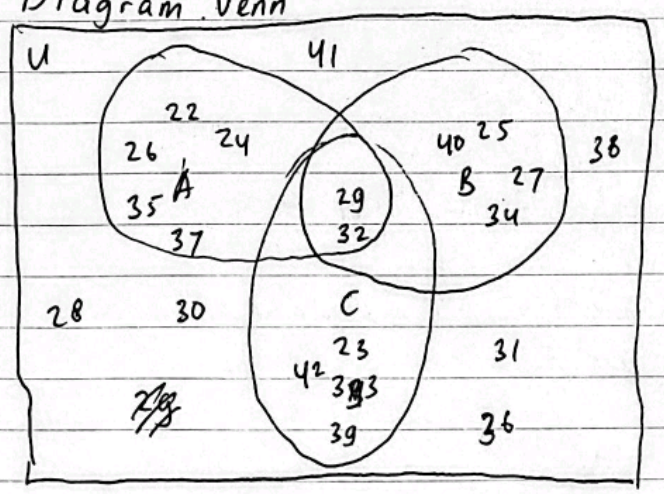
(Pelanggan dari referral)

$C = \{N+1, N+7, N+10, N+11, N+17, N+20\} \Rightarrow \{23, 29, 32, 33, 39, 42\}$

(pelanggan dari loyalty program)

$U = \{N, N+1, N+2, \dots, N+20\} \Rightarrow \{22, 23, 24, \dots, 42\}$

(a.) Diagram venn



- $A \cap B \cap C = \{29, 32\}$
- $A \cup B \cup C = \{22, 23, 24, 25, 26, 27, 29, 32, 33, 34, 35, 37, 39, 40, 42\}$
- $(A \cup B) \cap C'$

$A \cup B = \{22, 24, 25, 26, 27, 29, 32, 34, 35, 37, 40\}$

$C' \neq \{29, 32\}$

$\Rightarrow \{22, 24, 25, 26, 27, 34, 35, 37, 40\}$

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- $|A \oplus B| = (A \cup B) - (A \cap B)$

$(A \cup B) = \{22, 24, 25, 26, 27, 29, 32, 34, 35, 37, 40\}$

$(A \cap B) = \{29, 32\}$

=> Maka $|A \oplus B| = 22, 24, 25, 26, 27, 34, 35, 37, 40$

(c) Diberikan dua fungsi

$f(x) = k \cdot x + (a+b)$ (~~harga ucran~~)

$\Rightarrow 5x + 4$

$g(x) = \frac{x^2}{(b+1)} + a \Rightarrow \frac{x^2}{3} + 2$

- $(f \circ g)(x) = f(g(x))$

$= 5\left(\frac{x^2}{3} + 2\right) + 4$

$= \frac{5x^2}{3} + 10 + 4 = \frac{5x^2}{3} + 14$

- $(g \circ f)(x) = g(f(x))$

$= \frac{(5x+4)^2}{3} + 2$

$= \frac{25x^2 + 40x + 16}{3} + 2$

$= \frac{25x^2 + 40x + 16}{3} + \frac{6}{3} \Rightarrow \frac{25x^2 + 40x + 22}{33}$

Pembutian sifat komutatif dengan nilai $k = b+2 \Rightarrow x = 4$

$(f \circ g)(x) \Rightarrow (f \circ g)(4) = \frac{5(4)^2 + 14}{3}$

$= \frac{5 \cdot 16}{3} + \frac{42}{3}$

$= \frac{80}{3} + \frac{42}{3} = \frac{122}{3}$

$(g \circ f)(x) \Rightarrow (g \circ f)(4) = \frac{25(4)^2 + 40(4) + 22}{3} = \frac{25(16) + 160 + 22}{3}$

$= \frac{582}{3} = 194$

karena $\frac{122}{3} \neq 194$, maka $(f \circ g)(x) \neq (g \circ f)(x)$, dan fungsi harga terbukti tidak bersifat komutatif

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(d) Invers fungsi ($F^{-1}(x)$)

$F(x) = y \Rightarrow y = 5x + 4$

$5x = y - 4$

$x = \frac{y-4}{5} \Rightarrow f^{-1}(x) = \frac{y-4}{5}$ //

$\Rightarrow$ Rp. $(k \cdot 100 + a + b)$ ribu $\Rightarrow (5 \cdot 100 + 2 + 2)$ ribu

$= 504$ ribu

Untuk mengetahui volume pembelian (x), masukkan angka 504 ke dalam rumus invers

$\Rightarrow x = f^{-1}(504)$

$= \frac{504 - 4}{5} = 100$ // $\Rightarrow$ agar tercapai pendapatan 504 ribu harus terjual sebanyak 100 unit

(e.) Analisis Fungsi $F(x) = 5x + 4$

- Uji Injektif (satu-satu) $\Rightarrow$ jika $f(x_1) = f(x_2)$, maka wajib $x_1 = x_2$

$5x_1 + 4 = 5x_2 + 4$

$5x_1 = 5x_2$

$x_1 = x_2 \Rightarrow$ Terbukti

- Uji Surjektif (pada / onto) $\Rightarrow$ untuk setiap nilai $y \in R$, harus memiliki pasangan $x \in R$

$y = 5x + 4$

$5x = y - 4$

$x = \frac{y-4}{5} \Rightarrow$ untuk setiap angka real y yang disubstitusikan, hasil perhitungan pasti berupa bilangan real (terbukti)

- karena f(x) injektif dan surjektif, maka otomatis f(x) bijektif.

Analisis fungsi $g(x) = \frac{x^2}{3} + 2$

- uji injektif

$\frac{x_1^2}{3} + 2 = \frac{x_2^2}{3} + 2$

$x_1^2 = x_2^2$

$x_1 = \pm x_2 \rightarrow$ karena x_1 bisa berupa positif atau negatif x_2 terbukti tidak injektif

misal

$g(3) = \frac{3^2}{3} + 2 = \frac{9}{3} + 2 = 5$ dua input menghasilkan satu output

$g(-3) = \frac{-3^2}{3} + 2 = \frac{9}{3} + 2 = 5$

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- Uji Surjektif

nilai kuadrat (x^2) pasti ≥ 0

nilai paling minimum $g(x) = 2$

=> uji dengan $y = 0$ => $0 = \frac{x^2}{3} + 2$

$-2 = \frac{x^2}{3}$ => $x^2 = -6$

terbukti tidak surjektif

- karena $g(x)$ tidak injektif dan surjektif, maka $g(x)$ tidak bijektif

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